Alireza Daneshvar

Optimization & Controls Engineer | Data Scientist

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Professional Summary

Optimization and controls engineer with an M.A.Sc. in building engineering, focused on energy storage dispatch and building energy systems. I translate operational problems into mathematical programs and constrained control policies using mathematical programming, optimization, model predictive control, differentiable predictive control, and reinforcement learning. I write production code in Python, SQL, and PySpark, build optimization models with Pyomo and Gurobi, and track experiments in MLflow. Research at Johns Hopkins University and Concordia University has trained me to work independently on open-ended problems, from framing the model to delivering a tested result.

Work Experience

Concordia University — **Montreal, Quebec, Canada**
Machine Learning Researcher — *Oct 2025 – Present*

- Build probabilistic models of well leakage risk from noisy time-series records with a hurdle (classify, then regress) model.
- Define validation protocols against historical records that expose gaps in the underlying data; presented at OMAE 2026.

Johns Hopkins University — **Remote, United States**
Machine Learning Researcher — *Feb 2025 – Present*

- Develop differentiable predictive control (DPC) methods that turn constrained optimization problems into control policies evaluated in a single forward pass, enabling fast real-time decisions.
- Design sparse, interpretable control policies with safe online adaptation, trained end to end through physics-based dynamics.
- Build benchmarks comparing MPC and learning-based controllers on closed-loop performance and constraint satisfaction.
- Contribute differentiable optimization components to NeuroMANCER (PNNL) via pull requests and code review.

Concordia University — **Montreal, Quebec, Canada**
Research Assistant — *May 2021 – Apr 2023*

- Formulated energy hub dispatch with compressed air energy storage as a constrained scheduling problem that minimizes operating cost, solved with a safe deep RL method combining primal-dual optimization and imitation learning.
- Built regression surrogates that linearize the storage system's thermodynamics to keep long-horizon optimization tractable.
- Benchmarked against MILP and deep RL, within 2.7% of optimal cost; published in the Journal of Energy Storage.

IranBar — **Tehran, Iran**
Data Scientist — *Jun 2019 – May 2020*

- Shipped price-prediction models into production, cutting quotation time 30% and raising agreement rates 25%.
- Built data pipelines with PySpark, pandas, and SQL to pre-process large-scale freight datasets for model training.
- Maintained deployed models with versioning, CI/CD, monitoring, and scheduled retraining.

Technical Skills

Optimization & Control: Mathematical Programming, MILP, MPC, Differentiable Predictive Control (DPC), Safe Reinforcement Learning
Solvers & Modeling: Pyomo, Gurobi, NeuroMANCER, EnergyPlus
Machine Learning: Time-Series Forecasting, Deep Learning, System Identification, Transformers
Programming & Data: Python, SQL, PySpark, Azure Databricks, pandas, NumPy, PyTorch, scikit-learn, Git
Experiment Tracking & Deployment: MLflow, Docker, FastAPI, CI/CD, AWS, PostgreSQL, Snowflake

Education

Concordia University — **Montreal, Quebec, Canada**
M.A.Sc. in Building Engineering (GPA 4.1/4.3, Outstanding Thesis) — *Jan 2021 – Apr 2023*

Iran University of Science and Technology — **Tehran, Iran**
B.A.Sc. in Mechanical Engineering (Ranked 5th in cohort) — *Sep 2016 – Aug 2020*

Publications

- A. Daneshvar Garmroodi and J. Drgona. **"SD-DPC: Sparse Dictionary Differentiable Predictive Control."** Submitted to the *American Control Conference (ACC)*, 2027. arXiv:2610.02466
- T. Banker, A. Daneshvar Garmroodi, N. Lawrence, J. Drgona, and A. Mesbah. **"Learning Safe Control Policies: Reinforcement Learning-Enabled Differentiable Predictive Control."** *IFAC World Congress*, 2026.
- A. Daneshvar Garmroodi, F. Nasiri, and F. Haghighat. **"Optimal Dispatch of an Energy Hub with Compressed Air Energy Storage: A Safe Reinforcement Learning Approach."** *Journal of Energy Storage*, 2022. Link

Relevant Projects

Operational Cost Optimization of the Dompark Complex Energy System

Python, Pyomo, Mathematical Programming

- Translated the energy supply problem of the complex into a mathematical program: decision variables for each unit's output, energy-balance constraints for the electrical, heating, and cooling loads, and equipment capacity limits.
- Modeled several candidate energy system configurations and optimized the operating cost of each over a full year of load data in Pyomo.
- Identified the configuration with the lowest annual operating cost, giving a quantitative basis for choosing how to supply the complex.

Optimal HVAC Control With Reinforcement Learning and Forecast Inputs

Python, PyTorch, OpenAI Gym, EnergyPlus, LSTM, DQN

- Modeled the building and its HVAC system in EnergyPlus and wrapped the simulation as an OpenAI Gym environment for closed-loop control.
- Trained an LSTM to forecast ambient air temperature and solar irradiance, and passed the forecasts to the controller as inputs.
- Trained a DQN agent to set the supply air flow rate, trading off energy consumption against indoor temperature requirements.
- Cut energy use by 10% compared with a conventional rule-based controller while keeping indoor temperature in range.

Fault Detection and Diagnosis of an HVAC Air Handling Unit

Python, scikit-learn, pandas

- Framed fault detection and diagnosis for an air handling unit as a supervised classification problem on real operating data from a single-zone HVAC system.
- Cleaned and pre-processed the sensor data, and used exploratory visualization to understand how faulty and normal operation differ.
- Trained and compared Decision Tree, Random Forest, SVM, KNN, and Gaussian Naive Bayes classifiers, validating each with k-fold cross-validation and assessing them with ROC curves and AUC.
- Identified the best-performing classifier for detecting air handling unit faults on held-out folds.

Building Control With Safe RL and Differentiable Predictive Control GitHub

Python, PyTorch, PD-DDPG, OpenAI Gym, NeuroMANCER

- Built a physics-based building thermal environment in OpenAI Gym with box constraints on indoor temperature and control inputs.
- Implemented primal-dual DDPG, which learns a Lagrange multiplier alongside the policy to enforce constraints, and a DPC policy trained by backpropagating through the differentiable building model over a prediction horizon.
- Evaluated both controllers on energy use and constraint violations; released the comparison as a worked example in NeuroMANCER.